ETH zürich

Plotting Wavefields from Research Module

Salome Bachmann

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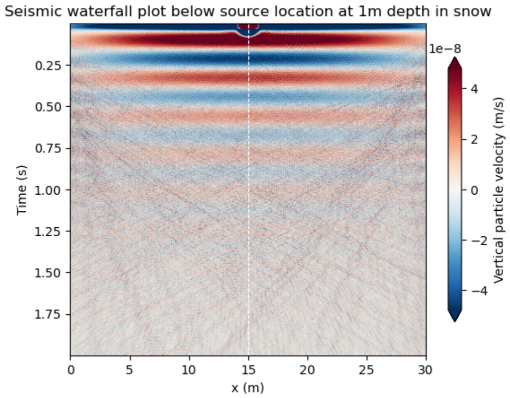
What have I been working on?

- Writing introduction
- Implementing feedback for introdcution
- Setup of remote machine
- Running simulations with lower source frequency
- Plotting waterfall plots, wiggle plots, single traces and frequency content

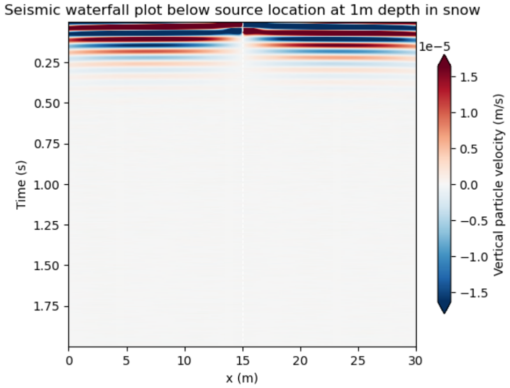
Simulation parameters used

- 30m by 3m domain
- one simulation with 150m by 3m domain
- 3 layers, all layers 1m thick
- absorbing boundaries at edges of domain
- 10Hz vector point source
- moment tensor source for skier (myy=0, mxx=0, mxy=3e4)

Wavefield Waterfall plots



(a) Vector point 2D source with $f_y = -1$. Source location indicated by white line. Colorbar shows particle velocity (m/s).

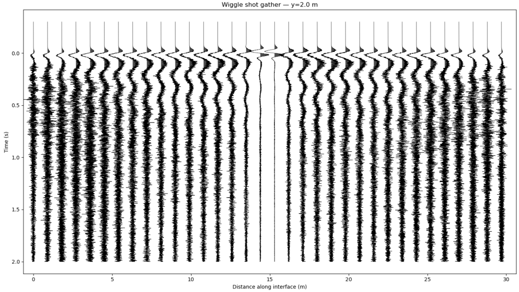


(b) Moment tensor source with $m_{xx} = 0$, $m_{yy} = 0$, $m_{xy} = 3 \times 10^4$. Source location indicated by white line. Colorbar shows particle velocity (m/s).

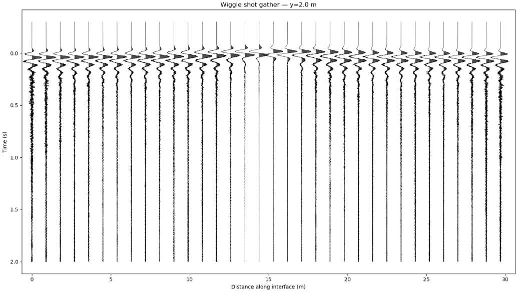
Waterfall plots interpretation

- Both sources show horizontal bands with high velocity: possible resonance between snow and air layer?
- Explained by tuning (Widess, 1973): possible constructive interference between top and bottom reflections of layer
- Vector point source shows some direct wave moveout (v-shaped pattern)
- Vector point source also shows some reflections at later times
- Problem of resonance could be solved by larger domain?

Wiggle Plots

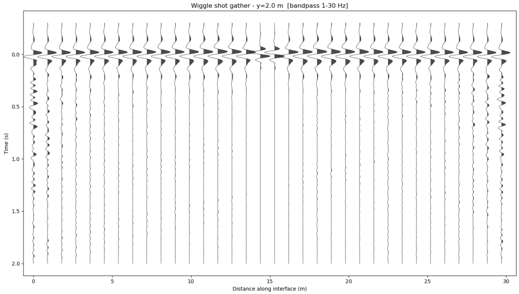


(a) Wiggle plot for vector point source.

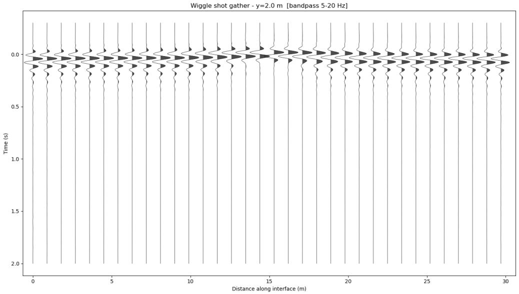


(b) Wiggle plot for moment tensor source.

Bandpass filtered wiggle plots



(a) Bandpass filtered wiggle plot for vector point source, Bandpass from 1 to 30HZ



(b) Bandpass filtered wiggle plot for moment tensor source, Bandpass from 5-20HZ

Wiggle Plots Interpretation

- Very noisy traces for vector point source, especially at the edges to the domain
- Bandpass filtering: Both moment tensor and vector point source show clear first breaks (large wiggles)
- SOme ground roll maybe (later arrivals)
- No hyperbolic reflections, which makes sense from domain setup

Single traces & Spectrograms

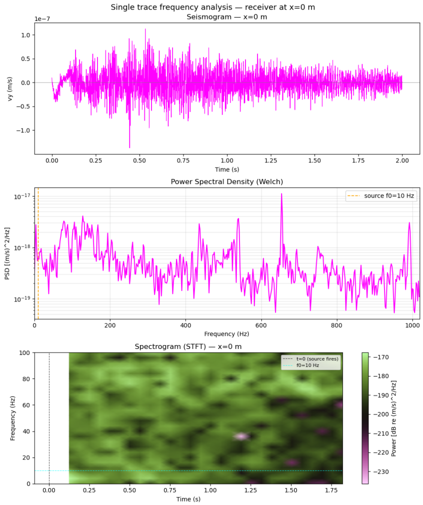


Figure: Single trace, power spectral density and spectrogram for vecror point source

Interpretation of single traces and spectrogram, power spectral density content

- Very noisy trace, source signal cannot be seen exactly (noise is stronger than signal)
- PSD shows energy at all frequencies: no source dominance
- Broadband signal after firing of source (indicated by black line)
- SOurce frequency shown by cyan lines in spectrogram
- Energy at all frequencies: field close to the source not dominated by source frequency
- Pink spot in spectrogram could correspond to peak in trace energy before 0.5s ?

Fixing the vector point source trace & PSD

- Single traces at other reciever locations show ricker pulse, but very noisy coda
- Bandpass filtering of traces like in wiggle plots?

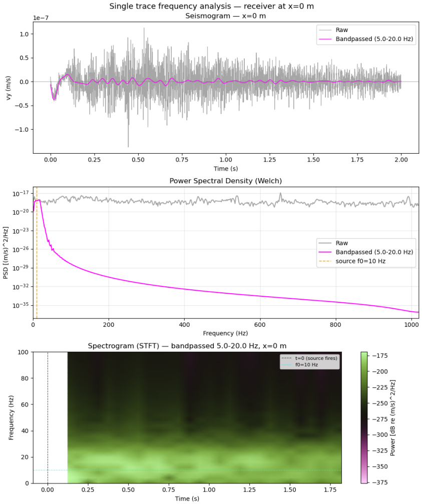


Figure: Bandpassed trace (pink, vs

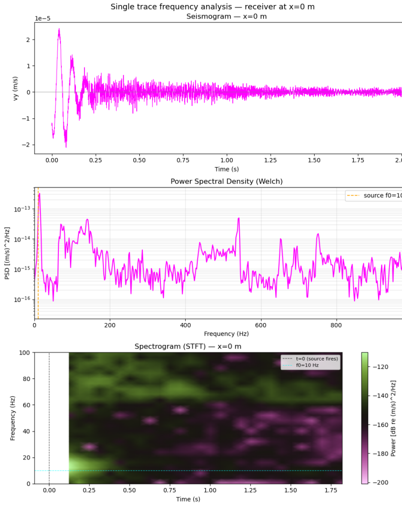


Figure: Single trace, power spectral density and spectrogram for moment tensor source

Interpretation of single traces and spectrogram, power spectral density content

- clear signal with decaying amplitide, way less noise than vector point source
- Bright green band in spectrogram: source frequency dominating near field
- Attenuation of green band (turns to black): energy disipation
- Pink spots: random reflections / scattering from interfaces?
- Peak in PSD around source frequency

Problems / Questions

- Simulations take very long to run on remote machine (2+ hours)
- Kernel keeps crashing for larger domain simulation
- Does it make sense to add topography to simulations?
- Does it make sense to add eg old snow layer (make the simulations multilayered)?

References

Widess, M. B. (1973).How thin is a thin bed? *Geophysics*, *38*(6), 1176–1180. https://doi.org/10.1190/1.1440403

Salome Bachmann
Master Student
sbachmann@student.ethz.ch

ETH Zurich
D-EAPS